

Nilesh Khadse

GIS Full Stack Developer



Experience

Senior Full Stack Engineer (GIS)

Deimos Space UK · Full-time

Jun 2021 - Present · 4 yrs 7 mos

Harwell Campus, England, United Kingdom

- Developed numerous Earth Observation applications, involving the creation of APIs, user interfaces, and processing large volumes of satellite data.
- Collaborated with the European Space Agency to successfully deliver several projects, including:
 - Designed a space debris collision risk assessment web application for satellite operators.
 - Implemented an Authentication and Access Management (IAM) service using Keycloak OpenID and an Accounting system for a microservices platform catering to Earth Observation application providers. This microservice plays a central role in the EU's Destination Earth (DestinE) initiative.

Skills: Python, docker and +4 skills

Senior MetOcean Software Developer

Fugro · Full-time

Nov 2019 - May 2021 · 1 yr 7 mos

Oxfordshire, England, United Kingdom

- Successfully implemented the Apache Airflow ETL platform for data processing workflows.
- Developed an atmospheric lightning data harvester in Python using the asyncio library, achieving a significant reduction in the server's CPU usage from 25% to 0.1% compared to its predecessor.
- Transformed the data ingestor application from a monolith Windows service to a containerized .NET Core application. This initiative not only reduced reliance on an aging data ingestion system but also made the team platform agnostic.
- Set up Elastic (ELK) analytics platform for monitoring workflows and applications, contributing to improved visibility and performance tracking.

Skills: Geoprocessing, Elastic Stack (ELK) and +3 skills

Principal Software Developer

Neftex Exploration Insights · Full-time

Jun 2014 - Oct 2019 · 5 yrs 5 mos

Oxford, United Kingdom

- Ideated and developed the 'QuickPlates' web mapping application, showcasing reconstructed data through geological times. This application also provided a client data processing service, marking a notable achievement.
- Devised a geological data reconstruction algorithm based on scientific literature, enabling the provision of 'reconstruction as a service' for client data.
- Developed a tool utilizing the OpenCV library to extract geographical contour features from historical map images.
- Built an aesthetic geological timescale using D3.js.
- Developed numerous ArcGIS plugins, web mapping applications and geospatial ETL workflows using FME.

Skills: React.js, .NET Framework and +3 skills

Senior GIS Analyst

HR Wallingford (SeaZone Solutions)

Apr 2013 - Jul 2014 · 1 yr 4 mos

SeaZone Solutions

4 yrs 7 mos

GIS Analyst

Jan 2009 - Mar 2013 · 4 yrs 3 mos

GIS Technician

Sep 2008 - Dec 2008 · 4 mos

Education

The University of Edinburgh

MSc, GIS

2007 - 2008

K. K. Wagh Institute of Engineering Education

B. Tech, Agricultural Engineering

2003 - 2007

Relevance to GDAC-SA Tender

Nilesh brings 16+ years of GIS and geospatial software development experience with proven expertise in building Earth Observation applications, web mapping platforms, and large-scale spatial data processing systems. His work with the European Space Agency on the DestinE initiative and development of geological reconstruction applications demonstrates capability directly applicable to the GDAC-SA Advanced Analytics Platform.

Key qualifications include:

- MSc in GIS from The University of Edinburgh
- Senior Full Stack Engineer (GIS) at Deimos Space UK collaborating with European Space Agency
- Developed Earth Observation applications involving APIs, user interfaces, and processing large volumes of satellite data
- Designed space debris collision risk assessment web application for satellite operators
- Implemented Authentication and Access Management (IAM) service using Keycloak OpenID for EU's Destination Earth (DestinE) initiative
- Built QuickPlates web mapping application for reconstructed geological data through geological time periods
- Devised geological data reconstruction algorithm based on scientific literature
- Developed numerous ArcGIS plugins, web mapping applications and geospatial ETL workflows using FME
- Implemented Apache Airflow ETL platform for data processing workflows
- Achieved 99.6% reduction in server CPU usage through optimized Python asyncio implementation
- Experience with Python, Docker, React.js, .NET Framework, ArcGIS, FME, D3.js, OpenCV, Elastic Stack (ELK)
- 16+ years combined experience in GIS analysis, software development, and geospatial systems (2008-present)